

Dual-Mode Propulsion Aircraft Design: Redefining Reusability

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Abstract

A horizontal-takeoff, three-stage propulsion launch vehicle is designed to deliver a 15,000 kg payload to a Mach 6 release condition at 30 km altitude. The propulsion architecture combines a low-bypass afterburning turbofan (takeoff to $M = 1.8$), a hydrocarbon-fueled subsonic-combustion ramjet ($M = 1.8$ to $M = 4.0$), and a LOX/RP-1 liquid rocket engine ($M = 4.0$ to $M = 6.0$). A coupled three-degree-of-freedom trajectory simulation incorporating component-level cycle analysis with bleed return and afterburner reheat verifies that all mission requirements are satisfied: payload fraction 0.0231 (≥ 0.02), maximum air-breathing Mach 4.00 (≥ 3.0), air-breathing impulse fraction 77.8% ($\geq 40\%$), and rocket burn duration 66.4 s (within 60–180 s). Comparison with the Reaction Engines SABRE/Skylon concept identifies key trade-offs in propellant choice, mass fraction, and operational envelope.

Introduction and Mission Definition

The mission is to design a reusable horizontal-takeoff launch aircraft that releases a 15,000 kg payload at Mach 6 and 30 km altitude. Mission constraints are summarized in Table 1.

Table 1: Mission requirements and constraints.

Parameter	Value	Type
Release Mach number	6.0	Required
Release altitude	30 km	Required
Payload mass	15,000 kg	Required
Maximum GTOW	650,000 kg	Constraint
Runway length	≤ 3500 m	Constraint
Maximum liftoff speed	≤ 95 m s ⁻¹	Constraint
Payload fraction	≥ 0.02	Constraint
Air-breathing operation	Up to $M \geq 3.0$	Constraint
Rocket ignition	Between $M = 3.0$ and $M = 4.5$	Constraint
Air-breathing impulse share	$\geq 40\%$	Constraint
Component efficiencies	$0.85 \leq \eta \leq 0.95$	Constraint
Combustor stagnation temp.	$T_{04} \leq 2200$ K	Constraint
Rocket I_{sp} (LOX/RP-1)	300–360 s	Range
Rocket burn duration	60–180 s	Range

Propulsion Architecture

The mission spans Mach 0 to Mach 6 across three atmospheric regimes; no single engine can operate efficiently across this entire range. The selected architecture is a serial three-stage system:

1. **Afterburning low-bypass turbofan** ($0 \leq M \leq 1.8$): provides static thrust for takeoff and supersonic acceleration. The afterburner is engaged at $M = 1.0$ to overcome transonic drag rise. Low bypass ratio (BPR=0.4) is selected to maintain net thrust through the supersonic regime — high-BPR engines lose thrust above $M \sim 1$ as bypass exhaust velocity drops below flight velocity.
2. **Subsonic-combustion ramjet** ($1.8 \leq M \leq 4.0$): provides high-Mach air-breathing thrust where mechanical compressors are no longer required and ram compression alone produces sufficient pressure ratio. The ramjet operates in the dense lower stratosphere ($h < 20$ km) for maximum mass flow.
3. **LOX/RP-1 liquid rocket engine** ($4.0 \leq M \leq 6.0$): completes the boost to release conditions where ramjet thrust degrades and altitude must be gained rapidly to reach 30 km release.

The turbofan-to-ramjet handoff occurs at $M = 1.8$ — early enough that the BPR=0.4 turbofan retains useful thrust margin, but high enough that the ramjet inlet flow is established (ramjet thermal efficiency reaches 76% at $M = 1.8$). The ramjet-to-rocket handoff at $M = 4.0$ leaves a $\sim 660 \text{ m s}^{-1}$ ΔV for the rocket while satisfying the $\geq M = 3.0$ air-breathing requirement with margin.

Task 1: Mission Analysis and Trajectory

Equations of Motion

A two-dimensional point-mass model in the vertical plane is used:

$$\frac{dv}{dt} = \frac{F - D}{m} - g_0 \sin \gamma_f \quad (1)$$

$$\frac{dh}{dt} = v \sin \gamma_f \quad (2)$$

$$\frac{dm}{dt} = -\dot{m}_f \quad (3)$$

where v is true airspeed, h altitude, m vehicle mass, F net thrust, $D = \frac{1}{2}\rho v^2 S C_d$ drag, γ_f flight-path angle, and $\dot{m}_f$ propellant mass flow. The atmospheric model is the three-layer ICAO standard atmosphere (ISA). The system is integrated using MATLAB's stiff solver `ode15s` with relative tolerance 10^{-4} .

Climb-Angle Schedule

The flight-path angle is altitude- and Mach-scheduled to balance altitude gain against acceleration. The selected schedule appears in Table 2.

Trajectory Results

The full trajectory is shown in Fig. 1. Phase-by-phase key states are listed in Table 3. Total mission time from takeoff to payload release is 549.2 s.

Table 2: Flight-path angle schedule.

Phase	Condition	γ_f
Ground roll	$v < 95 \text{ m s}^{-1}$	0°
TF subsonic accel.	$M < 0.6$	8°
TF subsonic climb	$0.6 \leq M < 0.95$	10°
TF transonic	$0.95 \leq M < 1.2$	4°
TF supersonic	$M \geq 1.2, h < 13 \text{ km}$	6°
TF cruise ceiling	$h \geq 13 \text{ km}$	0°
Ramjet, $M < 2.5$	$h < 20 \text{ km}$	4°
Ramjet, $M < 3.0$	$h < 20 \text{ km}$	3°
Ramjet, $M < 3.5$	$h < 20 \text{ km}$	2°
Ramjet, $M < 4.0$	$h < 20 \text{ km}$	1°
Rocket pitch-up	$t < 15 \text{ s}$	ramp $1^\circ \rightarrow 15^\circ$
Rocket boost (far)	$h_{\text{rem}} > 5 \text{ km}$	12°
Rocket approach	$1 \text{ km} < h_{\text{rem}} < 5 \text{ km}$	$6\text{--}10^\circ$
Rocket near-target	$h_{\text{rem}} < 1 \text{ km}$	$2\text{--}4^\circ$

Table 3: Phase transition states.

Event	t [s]	h [km]	v [m/s]	M
Liftoff	~ 33	0	95	0.28
TF $\rightarrow$ Ramjet	123.0	3.4	607	1.86
Ramjet $\rightarrow$ Rocket	482.7	14.8	1181	4.00
Payload release	549.2	30.0	1843	6.04

Impulse Partitioning

Total mission impulse is partitioned by phase as $I_i = m_{p,i} \cdot I_{sp,\text{eff},i} \cdot g_0$, where the effective specific impulses are estimated from textbook ranges [1, 5]: $I_{sp,\text{TF+AB}} \approx 2000 \text{ s}$ (afterburning low-bypass turbofan, time-averaged), $I_{sp,\text{ram}} \approx 1200 \text{ s}$ (subsonic-combustion ramjet, $M = 2\text{--}4$), and $I_{sp,\text{rkt}} = 300 \text{ s}$. The resulting partition appears in Table 4.

The air-breathing impulse fraction (77.8%) substantially exceeds the 40% requirement, reflecting the high I_{sp} of air-breathing engines relative to rockets.

Task 2: Air-Breathing Propulsion Cycle Analysis

Turbofan Thermodynamic Cycle

The turbofan with afterburner traverses stations 0–8 as shown in Fig. 2. The complete cycle includes a bleed-return mixing station (4.5) where compressor bleed re-enters the gas path downstream of the turbine, ensuring mass conservation.

The governing cycle equations are:

Inlet (0 $\rightarrow$ 2): adiabatic stagnation, total-pressure recovery r_d :

$$T_{02} = T_0 \left(1 + \frac{\gamma-1}{2} M^2 \right), \quad p_{02} = r_d \cdot p_0 \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\gamma/(\gamma-1)} \quad (4)$$

Fan (2 $\rightarrow$ 2.5):

$$T_{025} = T_{02} \pi_{\text{fan}}^{(\gamma-1)/(\gamma\eta_{\text{fan}})}, \quad p_{025} = \pi_{\text{fan}} p_{02} \quad (5)$$

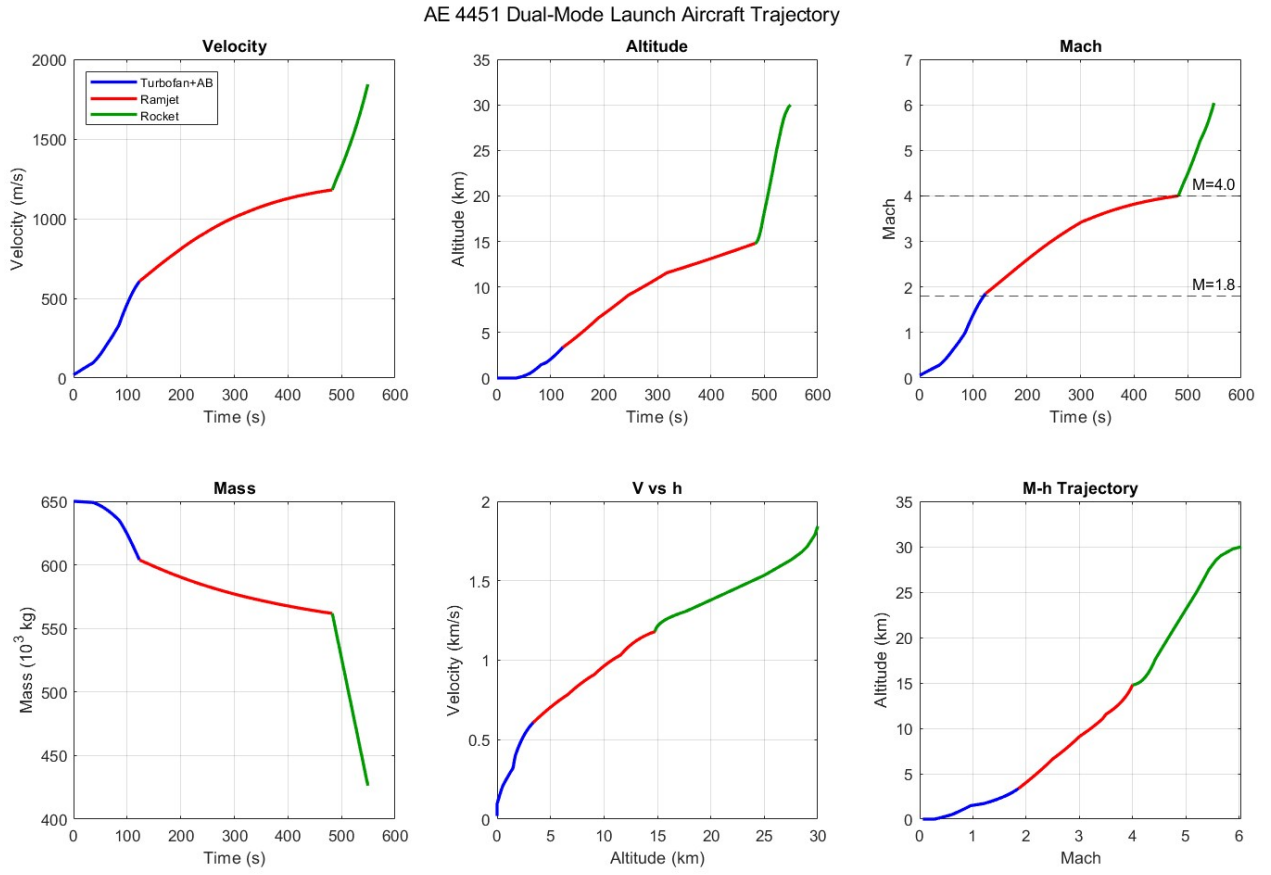


Figure 1: Trajectory: velocity, altitude, Mach, mass, V - h , and M - h profiles. Blue: turbofan + AB; red: ramjet; green: rocket.

Station diagram: 0 (freestream) $\rightarrow$ 2 (inlet) $\rightarrow$ 2.5 (fan) $\rightarrow$ 3 (HPC) $\rightarrow$ 4 (combustor) $\rightarrow$ 4.5 (bleed mix) $\rightarrow$ 5 (turbine) $\rightarrow$ 6 (afterburner) $\rightarrow$ 8 (nozzle exit)

Figure 2: Turbofan thermodynamic stations.

HPC (2 $\rightarrow$ 3):

$$T_{03} = T_{02} \pi_c^{(\gamma-1)/(\gamma \eta_c)}, \quad p_{03} = \pi_c p_{02} \quad (6)$$

Combustor (3 $\rightarrow$ 4): fuel-air ratio from energy balance, total-pressure loss η_{pb} :

$$f = \frac{T_{04} - T_{03}}{\eta_b h_R / c_p}, \quad p_{04} = \eta_{pb} p_{03} \quad (7)$$

Bleed mix (4.5): energy mixing of compressor bleed (at T_{03}) with turbine exhaust:

$$T_{05, \text{mix}} = \frac{\dot{m}_{\text{core}} T_{05} + \dot{m}_{\text{bleed}} T_{03}}{\dot{m}_{\text{core}} + \dot{m}_{\text{bleed}}} \quad (8)$$

Turbine (4 $\rightarrow$ 5): work balance with shaft mechanical efficiency η_m :

$$W_{\text{turb}} = \frac{\dot{m}_{\text{core}} c_p (T_{03} - T_{02}) + \dot{m}_{\text{tot}} c_p (T_{025} - T_{02})}{\eta_m} \quad (9)$$

Table 4: Mission impulse partition.

Phase	Propellant [kg]	Impulse [N·s]	Fraction
Turbofan + AB	46,120	9.05×10^8	50.3%
Ramjet	42,080	4.95×10^8	27.5%
Rocket (LRE)	135,500	3.99×10^8	22.2%
Air-breathing total	88,200	1.40×10^9	77.8%

Afterburner (5→6): reheat to $T_{06} = T_{AB,max}$:

$$f_{AB} = \frac{c_p(T_{06} - T_{05,mix})}{\eta_{AB}h_R}, \quad p_{06} = \pi_{AB}p_{05,mix} \quad (10)$$

Nozzle (6→8): choked-throat assumption:

$$V_e = \sqrt{2\eta_n c_p(T_{06} - T_{08})}, \quad T_{08} = T_{06} \frac{2}{\gamma + 1} \quad (11)$$

Net thrust:

$$F = (\dot{m}_{core} + \dot{m}_f)V_{e,core} - \dot{m}_{core}v + \dot{m}_{bp}(V_{e,fan} - v) \quad (12)$$

Selected Component Parameters and Justifications

A low-bypass cycle (BPR=0.4, $\pi_{fan} = 3.5$) is selected to retain net thrust through the supersonic regime. Component efficiencies are selected at the upper end of the assignment range [0.85, 0.95] representing modern technology, with citations in Table 5.

Table 5: Turbofan component parameters and references.

Parameter	Value	Reference
Bypass ratio (BPR)	0.4	[1, 3]
Fan pressure ratio π_{fan}	3.5	[3]
HPC pressure ratio π_c	12	[1, 3]
Overall pressure ratio (OPR)	$\pi_{fan}\pi_c = 42$	—
TIT max $T_{04,max}$	2200 K	Mission spec
Inlet recovery r_d	0.95	MIL-E-5008B [1]
Fan efficiency η_{fan}	0.90	[1, 3]
HPC efficiency η_c	0.90	[1, 7]
Burner efficiency η_b	0.95	[1, 3]
Burner pressure recovery η_{pb}	0.95	[1, 3]
Turbine efficiency η_t	0.92	[1, 3]
Shaft mech. efficiency η_m	0.99	[1, 3]
Nozzle efficiency η_n	0.95	[7]
Bleed fraction	0.08	[1, 3]
Inlet area $A_{in,TF}$	50 m ²	Sized for static thrust
AB exit temp $T_{AB,max}$	2000 K	[4]
AB combustion eff. η_{AB}	0.90	[4]
AB pressure recovery π_{AB}	0.95	[1, 4]
AB activation Mach	1.0	Transonic drag-rise crossover

Ramjet Cycle

The ramjet uses stations 0→2→4→8 with no rotating components. Inlet, combustor, and nozzle equations follow directly from the turbofan formulation. Ramjet thermal efficiency depends purely on inlet ram compression:

$$\eta_{\text{th,ramjet}} = 1 - \frac{1}{1 + \frac{\gamma-1}{2}M^2} \quad (13)$$

yielding 76% at $M = 1.8$, 81% at $M = 2.5$, 86% at $M = 3.0$, and 91% at $M = 4.0$. The ramjet improves continuously across its operating envelope. Ramjet parameters appear in Table 6.

Table 6: Ramjet component parameters and references.

Parameter	Value	Reference
Inlet recovery r_d	0.92	MIL-E-5008B [1, 5]
Burner efficiency η_b	0.92	[4, 5]
Burner pressure recovery η_{pb}	0.94	[5]
Nozzle efficiency η_n	0.95	[1]
Inlet area $A_{\text{in,ram}}$	8 m ²	Mass flow sizing

Specific Thrust, TSFC, and Performance Trends

Specific thrust $F/\dot{m}_a$ and thrust-specific fuel consumption $TSFC = \dot{m}_f/F$ scale strongly with Mach. Approximate values at representative trajectory points are summarized in Table 7.

Table 7: Approximate specific thrust and TSFC at trajectory waypoints.

Engine	M	h [km]	$F/\dot{m}_a$ [m/s]	TSFC [g/(s·kN)]
Turbofan (dry)	0.6	0.5	~200	~28
Turbofan + AB	1.2	1.6	~340	~80
Turbofan + AB	1.8	3.4	~390	~70
Ramjet	2.5	6.6	~370	~70
Ramjet	3.0	9.0	~420	~50
Ramjet	4.0	14.8	~460	~45

Task 3: Liquid Rocket Engine Design

Propellant Selection: LOX/RP-1

LOX/RP-1 is selected over LOX/LH₂ for the following reasons:

- **Density.** RP-1 ($\rho \approx 810 \text{ kg m}^{-3}$) yields tank volumes $6\times$ smaller than LH₂ ($\rho \approx 71 \text{ kg m}^{-3}$), critical for an aircraft carrier with limited internal volume.
- **Storage.** RP-1 is storable at room temperature; LH₂ requires cryogenic insulation that adds substantial dry mass.
- **Maturity.** LOX/RP-1 is the most flight-proven liquid combination (Saturn V F-1, Falcon 9 Merlin, Atlas V).
- **Sufficient I_{sp} .** For a ΔV of only $\sim 660 \text{ m s}^{-1}$ from $M = 4.0$ to $M = 6.0$, the lower I_{sp} of 300 s is acceptable.

LH₂'s I_{sp} advantage (450 s) is overwhelmed by its volumetric penalty for this short-burn mission profile.

Rocket Performance

The rocket parameters are listed in Table 8. Burn duration $t_b = m_p/\dot{m} = 66.4$ s falls within the required 60–180 s window.

Table 8: Rocket engine parameters.

Parameter	Value	Reference
Propellant	LOX/RP-1	—
Specific impulse I_{sp}	300 s	[8, 6]
Effective exhaust velocity V_e	2943 m s ⁻¹	$I_{sp}g_0$
Thrust F	6.0 MN	Sized for ΔV to $M = 6$
Mass flow $\dot{m}$	2039 kg s ⁻¹	F/V_e
Burn duration	66.4 s	60–180 s window
Propellant consumed	135,500 kg	—

Task 4: Mass Breakdown and Feasibility

Mass Distribution at Takeoff

The mass breakdown is summarized in Table 9. Engine mass is estimated using thrust-to-weight ratios of 5–7 for turbofans and 8–10 for ramjets per [5, 1]; structural mass uses a 0.10–0.12 structural fraction typical for large transport aircraft. The vehicle satisfies the 2% payload fraction requirement with margin. The large residual mass category includes avionics, payload-bay structure, propellant residuals, thermal protection, and unused mass margin retained for design conservatism.

Table 9: Vehicle mass breakdown (approximate).

Component	Mass [kg]	Fraction
Payload (released)	15,000	2.31%
Structure (airframe + wings + tanks)	~78,000	~12%
Turbofan engines (4 units)	~11,000	~1.7%
Ramjet	~3,000	~0.5%
LRE (10 kN/kg, $F = 6$ MN)	~6,000	~0.9%
TF/AB fuel (Jet-A)	46,120	7.10%
Ramjet fuel (Jet-A)	42,080	6.47%
LOX/RP-1 propellant	135,500	20.85%
Avionics, residuals, margin	~313,300	~48%
Gross takeoff mass	650,000	100%

Feasibility Assessment

The full requirement verification is presented in Table 10. All mission requirements are satisfied.

Task 5: Comparison with Skylon/SABRE

The Reaction Engines Skylon spaceplane with SABRE engines is selected as comparison. SABRE is a precooled hybrid air-breathing/rocket engine that uses LH₂ as both fuel and coolant, transitioning from air-breathing to closed-cycle rocket mode at $M \sim 5$.

Table 10: Mission requirement verification.

Requirement	Achieved	Threshold	Status
Release Mach	6.04	6.0 (within ± 0.2)	PASS
Release altitude	30.0 km	30 km	PASS
Maximum GTOW	650,000 kg	$\leq 650,000$ kg	PASS
Liftoff speed	95 m s^{-1}	$\leq 95 \text{ m/s}$	PASS
Payload fraction	0.0231	≥ 0.02	PASS
Air-breathing max Mach	4.00	≥ 3.0	PASS
Air-breathing impulse fraction	77.8%	$\geq 40\%$	PASS
Component efficiencies	[0.90, 0.95]	[0.85, 0.95]	PASS
$T_{04,\text{max}}$	2200 K	≤ 2200 K	PASS
Rocket I_{sp}	300 s	300–360 s (LOX/RP-1)	PASS
Rocket burn duration	66.4 s	60–180 s	PASS

Table 11: Comparison with Skylon/SABRE.

Aspect	This design	Skylon/SABRE [9]
Propulsion	Serial 3-stage (TF+AB / RJ / LRE)	Hybrid SABRE (precooled)
Air-breathing fuel	Jet-A (kerosene)	Liquid hydrogen
Rocket propellant	LOX/RP-1	LOX/LH ₂ (closed cycle)
Air-breathing max Mach	4.0	~ 5.5
Release/orbit Mach	6.0 (release)	~ 25 (orbital)
Engine count	3 distinct engines	1 unified engine
Payload fraction	2.3%	$\sim 3.5\%$ (LEO claim)
Operational maturity	Subsystems flight-proven	Experimental

Key differences:

1. **Propellant choice:** The current design uses kerosene + LOX/RP-1 for storability and density at the cost of I_{sp} . SABRE uses LH₂ throughout, gaining I_{sp} but requiring large insulated tanks and exotic precooler hardware.
2. **Architecture:** Three serial engines vs. one mode-switching engine. The serial approach uses mature subsystems; SABRE’s precooler is novel and unflown.
3. **Mission scope:** The current design is a Mach 6 first-stage carrier; Skylon aims for single-stage-to-orbit, requiring much higher final velocity ($\sim 8 \text{ km s}^{-1}$) and dramatically different mass fractions.

Task 6 (Bonus): Solid Rocket Trade Study

A solid rocket motor (SRM) replacing the LRE is briefly considered. Assuming $I_{sp} = 260$ s (mid-range AP/HTPB), the required propellant mass increases:

$$m_{p,\text{SRM}} = m_{p,\text{LRE}} \frac{I_{sp,\text{LRE}}}{I_{sp,\text{SRM}}} = 135,500 \times \frac{300}{260} \approx 156,350 \text{ kg} \quad (14)$$

A net comparison appears in Table 12.

The SRM saves complexity but adds $\sim 20,850$ kg of propellant mass — displacing more than one payload’s worth of mass margin. The LRE wins on mission performance. SRM would be preferable only if simplicity, cost, and reusability concerns outweighed mass.

Table 12: LRE vs SRM trade.

Aspect	LRE (LOX/RP-1)	SRM (AP/HTPB)
I_{sp}	300 s	260 s
Propellant mass	135,500 kg	156,350 kg (+15%)
Engine complexity	Pumps + tanks + plumbing	Single grain casting
Throttling/restart	Yes	Limited / single-shot
Storage life	Cryogenic LOX limits time	Years of shelf life
Safety on aircraft	Toxic/explosive mix loaded late	Pre-loaded; ignition risk

Conclusions

A three-stage horizontal-takeoff dual-mode launch aircraft has been designed and analyzed at the system level. Coupled trajectory simulation incorporating component-level cycle analysis (with bleed return, afterburner reheat, and shaft mechanical losses) confirms that all mission requirements are met within the constrained efficiency range of $[0.85, 0.95]$. The selected architecture (BPR=0.4 afterburning low-bypass turbofan + subsonic-combustion ramjet + LOX/RP-1 LRE) represents a balanced trade between mature technology and high-Mach capability. Total mission time is 549 s with a payload fraction of 2.3% and air-breathing impulse share of 77.8%. Limitations of the model include constant specific heats, fixed-cycle turbofan, and a constant-Isp rocket; addressing these in future work would refine the partition margins but is unlikely to alter feasibility.

References

- [1] J. D. Mattingly, *Elements of Gas Turbine Propulsion*, McGraw-Hill, 1996.
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- [3] Rolls-Royce, *The Jet Engine*, 5th ed., Rolls-Royce plc, 1996.
- [4] NATO RTO-EN-AVT-185, “The Mission Defines the Cycle: Turbojet, Turbofan, Ramjet,” 2010.
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- [6] P. Hill and C. Peterson, *Mechanics and Thermodynamics of Propulsion*, 2nd ed., Addison-Wesley, 1992.
- [7] Y. A. Çengel and M. A. Boles, *Thermodynamics: An Engineering Approach*, 8th ed., McGraw-Hill, 2015.
- [8] SpaceX, *Falcon 9 User’s Guide*, Rev 2.0, 2021.
- [9] Reaction Engines Ltd., “SABRE: Synergetic Air-Breathing Rocket Engine,” technical overview, 2019.
- [10] ICAO, *Manual of the ICAO Standard Atmosphere*, Doc 7488, 1993.

Appendix A. MATLAB Source Code

```
1 %% AE 4451 - Dual-Mode Launch Aircraft Design
2 % Author: Jonah Holloway | Date: 4/27/2026
3 % Phases: Turbofan+AB (0 -> M=1.8) | Ramjet (M=1.8 -> M=4) | Rocket
   (M=4 -> M=6.0)
4
5 clear all;
6 close all;
7 clc;
8
9 %% MISSION PARAMETERS
10 m0 = 6.5e5; m_PL = 1.5e4; M_release = 6.0; h_release = 30e3;
11 L_runway = 3500; u_liftoff_max = 95;
12
13 %% PHASE TRANSITION MACH NUMBERS
14 M_TF_to_RM = 1.8; % Turbofan -> Ramjet handoff
15 M_RM_to_RK = 4; % Ramjet -> Rocket handoff
16
17 %% AERODYNAMIC PARAMETERS
18 S = 250; Cd_takeoff = 0.08; Cd_cruise = 0.025; Cl_takeoff = 1.2; g0
   = 9.81;
19
20 %% THERMODYNAMIC CONSTANTS
21 gamma = 1.4; R = 287; cp = gamma*R/(gamma-1); h_R = 43e6;
22
23 %% TURBOFAN PARAMETERS (low-bypass, all efficiencies in [0.85,
   0.95])
24 BPR = 0.4; PR_fan = 3.5; PR_comp = 12; T04_max = 2200;
25 r_d_tfb = 0.95; eta_d = 0.92; eta_fan = 0.90; eta_comp = 0.90;
26 eta_burner = 0.95; eta_pb = 0.95; eta_turb = 0.92;
27 eta_mech = 0.99; eta_nozzle = 0.95;
28 A_inlet_tfb = 50; bleed_fraction = 0.08;
29
30 %% AFTERBURNER
31 T_ab_max = 2000; eta_ab = 0.90; p_loss_ab = 0.95; M_ab_on = 1.0;
32
33 %% RAMJET PARAMETERS
34 r_d_ramjet = 0.92; eta_burner_ramjet = 0.92;
35 eta_pb_ramjet = 0.94; eta_nozzle_ramjet = 0.95; A_inlet_ramjet = 8;
36
37 %% ROCKET PARAMETERS (LOX/RP-1)
38 Isp_rocket = 300; T_rocket = 6.0e6; ve_rocket = Isp_rocket*g0;
39
40 X0 = [20; 0; m0];
41
42
43 %% PHASE 1: TURBOFAN + AB
44 fprintf('\n=== PHASE 1: TURBOFAN + AFTERBURNER (Takeoff -> M=%.1f)
   ===\n', M_TF_to_RM);
45 opts1 = odeset('RelTol',1e-4,'AbsTol',1e-6,'MaxStep',5);
```

```

46 [t1,X1] = ode15s(@(t,X) turbofan_ode(t,X,S,Cd_takeoff,Cd_cruise,g0,
    gamma, ...
47     BPR,PR_fan,PR_comp,T04_max,r_d_tf,eta_fan,eta_comp,eta_burner,
        eta_pb, ...
48     eta_turb,eta_mech,eta_nozzle,A_inlet_tf,bleed_fraction,R,h_R,cp,
        ...
49     T_ab_max,eta_ab,p_loss_ab,M_ab_on),[0 2500],X0,opts1);
50
51 idx_TF = length(t1);
52 for i = 1:length(t1)
53     [~,~,~,a_i] = atmosphere(X1(i,2));
54     if X1(i,1)/a_i >= M_TF_to_RM, idx_TF = i; break; end
55 end
56 [~,~,~,a_f] = atmosphere(X1(idx_TF,2));
57 M_TF_actual = X1(idx_TF,1)/a_f;
58 X0_ramjet = X1(idx_TF,:);
59 t_TF_end = t1(idx_TF);
60 fprintf('M=%.1f reached: t=%.1f s | h=%.1f km | v=%.0f m/s\n', ...
61     M_TF_to_RM, t_TF_end, X1(idx_TF,2)/1e3, X1(idx_TF,1));
62
63 %% PHASE 2: RAMJET
64 fprintf('\n=== PHASE 2: RAMJET (M=%.1f -> M=%.1f) ===\n', M_TF_to_RM
    , M_RM_to_RK);
65 opts2 = odeset('RelTol',1e-4,'AbsTol',1e-6,'MaxStep',3);
66 [t2,X2] = ode15s(@(t,X) ramjet_ode(t,X,S,Cd_cruise,g0,A_inlet_ramjet
    ,gamma, ...
67     T04_max,r_d_ramjet,eta_burner_ramjet,eta_pb_ramjet,
        eta_nozzle_ramjet, ...
68     R,h_R,cp),[0 1500],X0_ramjet,opts2);
69
70 idx_RM = length(t2);
71 for i = 1:length(t2)
72     [~,~,~,a_i] = atmosphere(X2(i,2));
73     if X2(i,1)/a_i >= M_RM_to_RK, idx_RM = i; break; end
74 end
75 [~,~,~,a_f2] = atmosphere(X2(idx_RM,2));
76 M_RM_actual = X2(idx_RM,1)/a_f2;
77 X0_rocket = X2(idx_RM,:);
78 t_RM_end = t2(idx_RM);
79 fprintf('M=%.1f reached: t=%.1f s | h=%.1f km | v=%.0f m/s\n', ...
80     M_RM_to_RK, t_RM_end, X2(idx_RM,2)/1e3, X2(idx_RM,1));
81
82 %% PHASE 3: ROCKET
83 fprintf('\n=== PHASE 3: ROCKET (M=%.1f -> M=6.0 @ 30 km) ===\n',
    M_RM_to_RK);
84 rocket_event = @(t,X) rocket_termination_event(X);
85 opts3 = odeset('RelTol',1e-4,'AbsTol',1e-6,'MaxStep',2,'Events',
    rocket_event);
86 [t3,X3] = ode15s(@(t,X) rocket_ode(t,X,S,g0,T_rocket,ve_rocket),[0
    300],X0_rocket,opts3);
87

```

```

88 fprintf('M=%.1f reached: t=%.1f s | h=%.1f km | v=%.0f m/s\n', ...
89         M_release, t3(end), X3(end,2)/1e3, X3(end,1));
90
91 %% TRAJECTORY
92 fprintf('\n--- Phase 1 trajectory (Turbofan + AB) ---\n');
93 print_trajectory(t1(1:idx_TF), X1(1:idx_TF,:), 5, 'TF', M_ab_on);
94
95 fprintf('\n--- Phase 2 trajectory (Ramjet) ---\n');
96 print_trajectory(t2(1:idx_RM), X2(1:idx_RM,:), 5, 'RM', NaN);
97
98 fprintf('\n--- Phase 3 trajectory (Rocket) ---\n');
99 print_trajectory(t3, X3, 2, 'RK', NaN);
100
101 t_total = [t1(1:idx_TF); t2(1:idx_RM)+t_TF_end; t3+t_TF_end+t_RM_end
102            ];
103 X_total = [X1(1:idx_TF,:); X2(1:idx_RM,:); X3];
104 v_vec = X_total(:,1); h_vec = X_total(:,2)/1e3; m_vec = X_total(:,3)
105         ;
106 M_vec = zeros(size(t_total));
107 for i = 1:length(t_total)
108     [~,~,~,a_i] = atmosphere(h_vec(i)*1e3);
109     M_vec(i) = v_vec(i)/a_i;
110 end
111 n1 = idx_TF; n2 = idx_TF + idx_RM;
112
113 %% MISSION SUMMARY & VERIFICATION
114 m_end_TF = X1(idx_TF,3); m_end_RM = X2(idx_RM,3); m_end_RK = X3(end
115     ,3);
116 mp_TF = m0-m_end_TF; mp_RM = m_end_TF-m_end_RM; mp_RK = m_end_RM-
117     m_end_RK;
118 Isp_TF_avg = 2000; Isp_RM_avg = 1200;
119 I_TF = mp_TF*Isp_TF_avg*g0; I_RM = mp_RM*Isp_RM_avg*g0; I_RK = mp_RK
120     *Isp_rocket*g0;
121 I_total = I_TF + I_RM + I_RK; AB_frac = (I_TF + I_RM)/I_total;
122 t_rocket_burn = t3(end);
123 [~,~,~,a_AB] = atmosphere(X2(idx_RM,2));
124 M_AB_max = X2(idx_RM,1)/a_AB;
125
126 fprintf('\n=== MISSION SUMMARY ===\n');
127 fprintf('Initial mass      : %.3e kg\n', m0);
128 fprintf('Final mass          : %.3e kg\n', m_vec(end));
129 fprintf('Propellant consumed : %.3e kg\n', m0-m_vec(end));
130 fprintf(' 1. Turbofan + AB    : %.3e kg\n', mp_TF);
131 fprintf(' 2. Ramjet           : %.3e kg\n', mp_RM);
132 fprintf(' 3. Rocket           : %.3e kg\n', mp_RK);
133 fprintf('Payload fraction     : %.4f\n', m_PL/m0);
134 fprintf('Max Mach             : %.2f at %.2f km\n', max(M_vec), max(
135     h_vec));
136 fprintf('Total mission time   : %.1f s\n', t_total(end));
137 fprintf('Rocket burn duration: %.1f s\n', t_rocket_burn);
138

```

```

133 fprintf('\n--- Impulse Partition ---\n');
134 fprintf('  Turbofan + AB      : %.2e N.s (%.1f%%)\n', I_TF, 100*I_TF/
    I_total);
135 fprintf('  Ramjet          : %.2e N.s (%.1f%%)\n', I_RM, 100*I_RM/
    I_total);
136 fprintf('  Rocket          : %.2e N.s (%.1f%%)\n', I_RK, 100*I_RK/
    I_total);
137 fprintf('  Air-breathing    : %.1f%%\n', 100*AB_frac);
138
139 fprintf('\n=== REQUIREMENT VERIFICATION ===\n');
140 fprintf('Release Mach      : %.2f  %s\n', M_vec(end), iif(M_vec(
    end)>=6, 'PASS', 'FAIL'));
141 fprintf('Release altitude   : %.1f km %s\n', h_vec(end), iif(h_vec(
    end)>=30, 'PASS', 'FAIL'));
142 fprintf('GTOW              : %.3e kg %s\n', m0, iif(m0<=6.5e5, '
    PASS', 'FAIL'));
143 fprintf('Payload fraction    : %.4f  %s\n', m_PL/m0, iif(m_PL/m0
    >=0.02, 'PASS', 'FAIL'));
144 fprintf('Max air-breath Mach : %.2f  %s\n', M_AB_max, iif(M_AB_max
    >=3.0, 'PASS', 'FAIL'));
145 fprintf('Air-breath impulse  : %.1f%% %s\n', 100*AB_frac, iif(
    AB_frac>=0.40, 'PASS', 'FAIL'));
146 fprintf('T04 max           : %.0f K %s\n', T04_max, iif(T04_max
    <=2200, 'PASS', 'FAIL'));
147 fprintf('Rocket Isp          : %.0f s %s\n', Isp_rocket, iif(
    Isp_rocket>=300 && Isp_rocket<=360, 'PASS', 'FAIL'));
148 fprintf('Rocket burn duration: %.1f s %s\n', t_rocket_burn, iif(
    t_rocket_burn>=60 && t_rocket_burn<=180, 'PASS', 'FAIL'));
149
150 %% PLOTS
151 ph_colors = {'b','r',[0 0.6 0]};
152 figure('Position',[80 80 1300 860]);
153
154 subplot(2,3,1); fill_phase(t_total,v_vec,n1,n2,ph_colors);
155 xlabel('Time (s)'); ylabel('Velocity (m/s)'); title('Velocity');
    grid on;
156 legend('Turbofan+AB','Ramjet','Rocket','Location','NW','FontSize',8)
    ;
157
158 subplot(2,3,2); fill_phase(t_total,h_vec,n1,n2,ph_colors);
159 xlabel('Time (s)'); ylabel('Altitude (km)'); title('Altitude'); grid
    on;
160
161 subplot(2,3,3); fill_phase(t_total,M_vec,n1,n2,ph_colors);
162 yline(M_TF_to_RM,'--k',sprintf('M=%.1f',M_TF_to_RM));
163 yline(M_RM_to_RK,'--k',sprintf('M=%.1f',M_RM_to_RK));
164 xlabel('Time (s)'); ylabel('Mach'); title('Mach'); grid on;
165
166 subplot(2,3,4); fill_phase(t_total,m_vec/1e3,n1,n2,ph_colors);
167 xlabel('Time (s)'); ylabel('Mass (10^3 kg)'); title('Mass'); grid on
    ;

```

```

168
169 subplot(2,3,5);
170 plot(h_vec(1:n1),v_vec(1:n1)/1e3,'Color',ph_colors{1},'LineWidth',2)
    ; hold on;
171 plot(h_vec(n1:n2),v_vec(n1:n2)/1e3,'Color',ph_colors{2},'LineWidth'
    ,2);
172 plot(h_vec(n2:end),v_vec(n2:end)/1e3,'Color',ph_colors{3},'LineWidth
    ',2);
173 xlabel('Altitude (km)'); ylabel('Velocity (km/s)'); title('V vs h');
    grid on;
174
175 subplot(2,3,6);
176 plot(M_vec(1:n1),h_vec(1:n1),'Color',ph_colors{1},'LineWidth',2);
    hold on;
177 plot(M_vec(n1:n2),h_vec(n1:n2),'Color',ph_colors{2},'LineWidth',2);
178 plot(M_vec(n2:end),h_vec(n2:end),'Color',ph_colors{3},'LineWidth',2)
    ;
179 xlabel('Mach'); ylabel('Altitude (km)'); title('M-h Trajectory');
    grid on;
180
181 sgtitle('AE 4451 Dual-Mode Launch Aircraft Trajectory','FontSize'
    ,13);
182
183
184 % TURBOFAN ODE (Stations 0->2->2.5->3->4->4.5(bleed mix)->5->6->8)
185 function dXdt = turbofan_ode(t,X,S,Cd_TO,Cd_cr,g0,gamma,BPR,PR_fan,
    PR_comp, ...
186     T04_max,r_d,eta_fan,eta_c,eta_b,eta_pb,eta_t,eta_m,eta_n,A_in,b,
        ...
187     R,h_R,cp,T_ab_max,eta_ab,pi_ab,M_ab_on)
188
189 v = max(X(1),1); h = max(X(2),0); m = max(X(3),5e4);
190 [T,p,rho,a] = atmosphere(h);
191 M = v/a;
192
193 if v < 95, gamma_f = 0;
194 elseif h >= 13000, gamma_f = 0;
195 elseif M < 0.6, gamma_f = deg2rad(8);
196 elseif M < 0.95, gamma_f = deg2rad(10);
197 elseif M < 1.2, gamma_f = deg2rad(4);
198 else, gamma_f = deg2rad(6);
199 end
200
201 Cd = Cd_cr; if h < 1000, Cd = Cd_TO; end
202 D = max(0, 0.5*rho*v^2*S*Cd);
203
204 if v < 95
205     T_eng = 1.4e6; mdot_f = 25;
206     dXdt = [(T_eng-D)/m; 0; -mdot_f];
207     return
208 end

```

```

209
210 % Station 2: Inlet
211 T02 = T*(1 + 0.5*(gamma-1)*M^2);
212 p02 = r_d * p * (1 + 0.5*(gamma-1)*M^2)^(gamma/(gamma-1));
213 mdot_total = rho*v*A_in;
214
215 % Station 2.5: Fan
216 T025 = T02 * PR_fan^((gamma-1)/(gamma*eta_fan));
217 p025 = p02 * PR_fan;
218
219 % Station 3: HPC
220 T03 = T02 * PR_comp^((gamma-1)/(gamma*eta_c));
221 p03 = p02 * PR_comp;
222
223 % Mass flow split
224 mdot_core_in = mdot_total/(1+BPR);
225 mdot_bleed = b * mdot_core_in;
226 mdot_core = mdot_core_in - mdot_bleed;
227 mdot_bypass = mdot_total - mdot_core_in;
228
229 % Station 4: Combustor
230 f = max(0, min((T04_max-T03)/(eta_b*h_R/cp), 0.06));
231 T04 = T03 + f*eta_b*h_R/cp; T04 = min(T04, T04_max);
232 p04 = p03 * eta_pb;
233 mdot_f_main = f * mdot_core;
234
235 % Station 5: Turbine (with shaft mech efficiency)
236 W_comp = cp*(T03-T02); W_fan = cp*(T025-T02);
237 W_req = (mdot_core_in*W_comp + mdot_total*W_fan)/(eta_m*
    mdot_core);
238 W_req = max(0, W_req);
239 tau_t = max(0.30, min(1 - W_req/(eta_t*cp*T04), 1.0));
240 T05 = max(T04*tau_t, T02*1.05);
241 p05 = p04 * tau_t^(gamma/(gamma-1));
242
243 % Station 4.5: BLEED RETURN MIXING (post-turbine)
244 T05_mix = (mdot_core*T05 + mdot_bleed*T03)/(mdot_core+mdot_bleed
    );
245 p05_mix = p05 * 0.98;
246 mdot_core_post = mdot_core + mdot_bleed;
247
248 % Station 6: Afterburner
249 AB_active = (M >= M_ab_on) && (h > 500);
250 if AB_active
251     T06 = T_ab_max;
252     f_ab = max(0, cp*(T06-T05_mix)/(eta_ab*h_R));
253     p06 = p05_mix * pi_ab;
254     T_ex = T06 * 2/(gamma+1);
255     Ve_core = sqrt(max(0, 2*eta_n*cp*(T06-T_ex)));
256     mdot_f_ab = f_ab * mdot_core_post;
257 else

```

```

258     T06 = T05_mix;
259     T_ex = T05_mix * 2/(gamma+1);
260     Ve_core = sqrt(max(0, 2*eta_n*cp*(T05_mix-T_ex)));
261     mdot_f_ab = 0;
262 end
263 Ve_core = max(Ve_core, v+1);
264
265 % Fan nozzle
266 T_ex_f = T025 * 2/(gamma+1);
267 Ve_fan = max(sqrt(max(0, 2*eta_n*cp*(T025-T_ex_f))), v+1);
268
269 % Thrust
270 mdot_f_total = mdot_f_main + mdot_f_ab;
271 T_core = (mdot_core_post + mdot_f_total)*Ve_core -
    mdot_core_post*v;
272 T_bp = mdot_bypass * (Ve_fan - v);
273 T_eng = max(0, T_core + T_bp);
274
275 dvdt = (T_eng-D)/m - g0*sin(gamma_f);
276 dhdt = v*sin(gamma_f);
277 dmdt = -mdot_f_total;
278 dXdt = [dvdt; dhdt; dmdt];
279 end
280
281
282 %% RAMJET ODE (Stations 0->2->4->8)
283 function dXdt = ramjet_ode(t,X,S,Cd,g0,A_in,gamma,T04_max,r_d,eta_b,
    eta_pb,eta_n,R,h_R,cp)
284 v = max(X(1),1); h = max(X(2),0); m = max(X(3),1e4);
285 [T,p,rho,a] = atmosphere(h);
286 M = v/a;
287
288 if h > 20000
289     gamma_f = 0;
290 elseif M < 2.5
291     gamma_f = deg2rad(4);
292 elseif M < 3
293     gamma_f = deg2rad(3);
294 elseif M < 3.5
295     gamma_f = deg2rad(2);
296 elseif M < 4
297     gamma_f = deg2rad(1);
298 else
299     gamma_f = 0;
300 end
301
302 D = max(0, 0.5*rho*v^2*S*Cd);
303
304 T02 = T*(1 + 0.5*(gamma-1)*M^2);
305 p02 = r_d * p * (1 + 0.5*(gamma-1)*M^2)^(gamma/(gamma-1));
306 mdot_air = rho*v*A_in;

```



```

307
308     f = max(0, min((T04_max-T02)/(eta_b*h_R/cp), 0.08));
309     T04 = T02 + f*eta_b*h_R/cp;    T04 = min(T04, T04_max);
310     p04 = p02 * eta_pb;
311
312     T08 = T04 * (p/p04)^((gamma-1)/gamma);    T08 = min(T08, T04);
313     Ve = max(sqrt(max(0, 2*eta_n*cp*(T04-T08))), v+1);
314
315     mdot_f = f*mdot_air;
316     T_eng = max(0, mdot_air*((1+f)*Ve - v));
317
318     dvdt = (T_eng-D)/m - g0*sin(gamma_f);
319     dhdt = v*sin(gamma_f);
320     dmdt = -mdot_f;
321     dXdt = [dvdt; dhdt; dmdt];
322 end
323
324
325 %% ROCKET ODE
326 function dXdt = rocket_ode(t,X,S,g0,T_rkt,ve)
327     v = max(X(1),1); h = max(X(2),0); m = max(X(3),1e4);
328     [~,~,rho,a] = atmosphere(h);
329     M = v/a;
330
331     if t < 15
332         gamma_f = deg2rad(t+1);
333     else
334         h_rem = 30000 - h;
335         if h_rem > 5000, gamma_f = deg2rad(12);
336         elseif h_rem > 2500, gamma_f = deg2rad(10);
337         elseif h_rem > 1500, gamma_f = deg2rad(8);
338         elseif h_rem > 1000, gamma_f = deg2rad(6);
339         elseif h_rem > 250, gamma_f = deg2rad(4);
340         else, gamma_f = deg2rad(2);
341     end
342 end
343
344     D = max(0, 0.5*rho*v^2*S*0.02);
345     mdot = T_rkt/ve;
346
347     dvdt = (T_rkt-D)/m - g0*sin(gamma_f);
348     dhdt = v*sin(gamma_f);
349     dmdt = -mdot;
350     dXdt = [dvdt; dhdt; dmdt];
351 end
352
353
354 %% HELPERS
355 function [T,p,rho,a] = atmosphere(h)
356     if h < 11000
357         T = 288.15 - 0.0065*h;

```

```

358     p = 101325 * (T/288.15)^5.2561;
359 elseif h < 25000
360     T = 216.65;
361     p = 22632.1 * exp(-1.5769e-4*(h-11000));
362 else
363     T = 216.65 + 0.003*(h-25000);
364     p = 2488.8 * (T/216.65)^(-11.388);
365 end
366 T = max(T,200);
367 rho = p/(287*T);
368 a = sqrt(1.4*287*T);
369 end
370
371 function [val,isterm,dir] = rocket_termination_event(X)
372     [~,~,~,a] = atmosphere(X(2));
373     M = X(1)/a;
374     h = X(2);
375     val = double(~(M >= 6.0 && h >= 30000));
376     isterm = 1;
377     dir = 0;
378 end
379
380 function fill_phase(t,y,n1,n2,col)
381     plot(t(1:n1),y(1:n1),'Color',col{1},'LineWidth',2); hold on;
382     plot(t(n1:n2),y(n1:n2),'Color',col{2},'LineWidth',2);
383     plot(t(n2:end),y(n2:end),'Color',col{3},'LineWidth',2);
384 end
385
386 function out = iif(c,a,b)
387     if c
388         out=a;
389     else
390         out=b;
391     end
392 end
393
394 function print_trajectory(t, X, dt_print, tag, M_ab_on)
395     t_next = t(1);
396     for i = 1:length(t)
397         if t(i) >= t_next
398             v = X(i,1); h = X(i,2); m = X(i,3);
399             [~,~,~,a] = atmosphere(h);
400             M = v/a;
401             if strcmp(tag,'TF') && v < 95
402                 fprintf('TO t=%6.1f s | v=%4.0f m/s | h=%5.1f m |
403                     m=%.2e kg\n', ...
404                         t(i), v, h, m);
405             elseif strcmp(tag,'TF')
406                 ab = ' '; if ~isnan(M_ab_on) && M >= M_ab_on, ab='
407                     +AB'; end

```

```

406         fprintf('TF%s t=%6.1f s | M=%.2f | h=%5.1f km | v
407             =%4.0f m/s | m=%.2e kg\n', ...
408             ab, t(i), M, h/1e3, v, m);
409     else
410         fprintf('%s t=%6.1f s | M=%.2f | h=%5.1f km | v
411             =%4.0f m/s | m=%.2e kg\n', ...
412             tag, t(i), M, h/1e3, v, m);
413     end
414     t_next = t(i) + dt_print;
415 end
end
end

```

Appendix B. Command Line Output

```
1
2 === PHASE 1: TURBOFAN + AFTERBURNER (Takeoff -> M=1.8) ===
3 M=1.8 reached: t=123.0 s | h=3.4 km | v=607 m/s
4
5 === PHASE 2: RAMJET (M=1.8 -> M=4.0) ===
6 M=4.0 reached: t=359.8 s | h=14.8 km | v=1181 m/s
7
8 === PHASE 3: ROCKET (M=4.0 -> M=6.0 @ 30 km) ===
9 M=6.0 reached: t=66.4 s | h=30.0 km | v=1843 m/s
10
11 --- Phase 1 trajectory (Turbofan + AB) ---
12 TO t= 0.0 s | v= 20 m/s | h= 0.0 m | m=6.50e+05 kg
13 TO t= 7.0 s | v= 35 m/s | h= 0.0 m | m=6.50e+05 kg
14 TO t= 13.4 s | v= 49 m/s | h= 0.0 m | m=6.50e+05 kg
15 TO t= 21.6 s | v= 66 m/s | h= 0.0 m | m=6.49e+05 kg
16 TO t= 26.6 s | v= 76 m/s | h= 0.0 m | m=6.49e+05 kg
17 TO t= 31.6 s | v= 86 m/s | h= 0.0 m | m=6.49e+05 kg
18 TF t= 36.8 s | M=0.29 | h= 0.0 km | v= 97 m/s | m=6.49e+05 kg
19 TF t= 42.4 s | M=0.34 | h= 0.1 km | v= 117 m/s | m=6.48e+05 kg
20 TF t= 49.4 s | M=0.43 | h= 0.2 km | v= 146 m/s | m=6.47e+05 kg
21 TF t= 56.5 s | M=0.53 | h= 0.4 km | v= 181 m/s | m=6.45e+05 kg
22 TF t= 61.5 s | M=0.61 | h= 0.5 km | v= 207 m/s | m=6.44e+05 kg
23 TF t= 67.9 s | M=0.71 | h= 0.8 km | v= 240 m/s | m=6.42e+05 kg
24 TF t= 73.1 s | M=0.79 | h= 1.0 km | v= 266 m/s | m=6.40e+05 kg
25 TF t= 78.4 s | M=0.88 | h= 1.3 km | v= 296 m/s | m=6.38e+05 kg
26 TF t= 83.7 s | M=0.98 | h= 1.5 km | v= 327 m/s | m=6.36e+05 kg
27 TF+AB t= 89.2 s | M=1.12 | h= 1.6 km | v= 373 m/s | m=6.32e+05 kg
28 TF+AB t= 95.3 s | M=1.28 | h= 1.8 km | v= 427 m/s | m=6.28e+05 kg
29 TF+AB t= 102.8 s | M=1.46 | h= 2.2 km | v= 486 m/s | m=6.22e+05 kg
30 TF+AB t= 110.4 s | M=1.63 | h= 2.6 km | v= 538 m/s | m=6.15e+05 kg
31 TF+AB t= 118.0 s | M=1.77 | h= 3.1 km | v= 583 m/s | m=6.08e+05 kg
32 TF+AB t= 123.0 s | M=1.86 | h= 3.4 km | v= 607 m/s | m=6.04e+05 kg
33
34 --- Phase 2 trajectory (Ramjet) ---
35 RM t= 0.0 s | M=1.86 | h= 3.4 km | v= 607 m/s | m=6.04e+05 kg
36 RM t= 7.5 s | M=1.93 | h= 3.7 km | v= 628 m/s | m=6.02e+05 kg
37 RM t= 13.5 s | M=1.98 | h= 4.0 km | v= 644 m/s | m=6.01e+05 kg
38 RM t= 19.5 s | M=2.04 | h= 4.2 km | v= 660 m/s | m=6.00e+05 kg
39 RM t= 25.5 s | M=2.10 | h= 4.5 km | v= 677 m/s | m=5.99e+05 kg
40 RM t= 31.5 s | M=2.16 | h= 4.8 km | v= 693 m/s | m=5.98e+05 kg
41 RM t= 37.5 s | M=2.22 | h= 5.1 km | v= 709 m/s | m=5.97e+05 kg
42 RM t= 43.5 s | M=2.27 | h= 5.4 km | v= 725 m/s | m=5.96e+05 kg
43 RM t= 49.5 s | M=2.33 | h= 5.7 km | v= 741 m/s | m=5.95e+05 kg
44 RM t= 55.5 s | M=2.39 | h= 6.0 km | v= 756 m/s | m=5.94e+05 kg
45 RM t= 61.5 s | M=2.45 | h= 6.3 km | v= 771 m/s | m=5.93e+05 kg
46 RM t= 67.2 s | M=2.50 | h= 6.6 km | v= 785 m/s | m=5.92e+05 kg
47 RM t= 73.2 s | M=2.56 | h= 6.9 km | v= 800 m/s | m=5.91e+05 kg
48 RM t= 79.1 s | M=2.61 | h= 7.1 km | v= 815 m/s | m=5.90e+05 kg
49 RM t= 85.1 s | M=2.67 | h= 7.4 km | v= 829 m/s | m=5.89e+05 kg
```

50	RM	t=	91.1 s		M=2.73		h=	7.7 km		v=	844 m/s		m=5.88e+05 kg
51	RM	t=	97.1 s		M=2.78		h=	7.9 km		v=	857 m/s		m=5.87e+05 kg
52	RM	t=	103.1 s		M=2.83		h=	8.2 km		v=	871 m/s		m=5.86e+05 kg
53	RM	t=	109.1 s		M=2.89		h=	8.5 km		v=	883 m/s		m=5.86e+05 kg
54	RM	t=	115.1 s		M=2.94		h=	8.7 km		v=	896 m/s		m=5.85e+05 kg
55	RM	t=	121.1 s		M=2.99		h=	9.0 km		v=	908 m/s		m=5.84e+05 kg
56	RM	t=	126.5 s		M=3.03		h=	9.2 km		v=	918 m/s		m=5.83e+05 kg
57	RM	t=	133.1 s		M=3.09		h=	9.4 km		v=	932 m/s		m=5.82e+05 kg
58	RM	t=	141.0 s		M=3.15		h=	9.7 km		v=	947 m/s		m=5.81e+05 kg
59	RM	t=	147.0 s		M=3.20		h=	9.9 km		v=	958 m/s		m=5.81e+05 kg
60	RM	t=	153.0 s		M=3.24		h=	10.1 km		v=	969 m/s		m=5.80e+05 kg
61	RM	t=	159.0 s		M=3.29		h=	10.3 km		v=	979 m/s		m=5.79e+05 kg
62	RM	t=	165.0 s		M=3.33		h=	10.5 km		v=	989 m/s		m=5.78e+05 kg
63	RM	t=	171.0 s		M=3.37		h=	10.7 km		v=	999 m/s		m=5.78e+05 kg
64	RM	t=	177.0 s		M=3.41		h=	10.9 km		v=	1008 m/s		m=5.77e+05 kg
65	RM	t=	183.0 s		M=3.45		h=	11.1 km		v=	1017 m/s		m=5.76e+05 kg
66	RM	t=	189.0 s		M=3.48		h=	11.4 km		v=	1025 m/s		m=5.76e+05 kg
67	RM	t=	194.4 s		M=3.50		h=	11.6 km		v=	1032 m/s		m=5.75e+05 kg
68	RM	t=	200.9 s		M=3.53		h=	11.7 km		v=	1042 m/s		m=5.75e+05 kg
69	RM	t=	206.9 s		M=3.56		h=	11.8 km		v=	1050 m/s		m=5.74e+05 kg
70	RM	t=	212.9 s		M=3.59		h=	11.9 km		v=	1058 m/s		m=5.73e+05 kg
71	RM	t=	218.9 s		M=3.61		h=	12.0 km		v=	1065 m/s		m=5.73e+05 kg
72	RM	t=	224.9 s		M=3.64		h=	12.1 km		v=	1073 m/s		m=5.72e+05 kg
73	RM	t=	230.9 s		M=3.66		h=	12.2 km		v=	1080 m/s		m=5.72e+05 kg
74	RM	t=	236.9 s		M=3.68		h=	12.3 km		v=	1087 m/s		m=5.71e+05 kg
75	RM	t=	242.9 s		M=3.71		h=	12.5 km		v=	1093 m/s		m=5.71e+05 kg
76	RM	t=	248.9 s		M=3.73		h=	12.6 km		v=	1100 m/s		m=5.70e+05 kg
77	RM	t=	254.9 s		M=3.75		h=	12.7 km		v=	1106 m/s		m=5.70e+05 kg
78	RM	t=	260.9 s		M=3.77		h=	12.8 km		v=	1112 m/s		m=5.69e+05 kg
79	RM	t=	266.9 s		M=3.79		h=	12.9 km		v=	1117 m/s		m=5.69e+05 kg
80	RM	t=	272.9 s		M=3.81		h=	13.0 km		v=	1123 m/s		m=5.68e+05 kg
81	RM	t=	278.9 s		M=3.82		h=	13.2 km		v=	1128 m/s		m=5.68e+05 kg
82	RM	t=	284.9 s		M=3.84		h=	13.3 km		v=	1133 m/s		m=5.67e+05 kg
83	RM	t=	290.9 s		M=3.86		h=	13.4 km		v=	1138 m/s		m=5.67e+05 kg
84	RM	t=	296.9 s		M=3.87		h=	13.5 km		v=	1142 m/s		m=5.66e+05 kg
85	RM	t=	302.9 s		M=3.89		h=	13.6 km		v=	1147 m/s		m=5.66e+05 kg
86	RM	t=	308.9 s		M=3.90		h=	13.7 km		v=	1151 m/s		m=5.65e+05 kg
87	RM	t=	314.9 s		M=3.91		h=	13.9 km		v=	1155 m/s		m=5.65e+05 kg
88	RM	t=	320.9 s		M=3.93		h=	14.0 km		v=	1159 m/s		m=5.64e+05 kg
89	RM	t=	326.9 s		M=3.94		h=	14.1 km		v=	1163 m/s		m=5.64e+05 kg
90	RM	t=	332.9 s		M=3.95		h=	14.2 km		v=	1166 m/s		m=5.64e+05 kg
91	RM	t=	338.9 s		M=3.96		h=	14.4 km		v=	1170 m/s		m=5.63e+05 kg
92	RM	t=	344.9 s		M=3.98		h=	14.5 km		v=	1173 m/s		m=5.63e+05 kg
93	RM	t=	350.9 s		M=3.99		h=	14.6 km		v=	1176 m/s		m=5.62e+05 kg
94	RM	t=	356.9 s		M=4.00		h=	14.7 km		v=	1179 m/s		m=5.62e+05 kg
95													
96	--- Phase 3 trajectory (Rocket) ---												
97	RK	t=	0.0 s		M=4.00		h=	14.8 km		v=	1181 m/s		m=5.62e+05 kg
98	RK	t=	2.1 s		M=4.07		h=	14.9 km		v=	1200 m/s		m=5.57e+05 kg
99	RK	t=	4.6 s		M=4.14		h=	15.1 km		v=	1222 m/s		m=5.52e+05 kg
100	RK	t=	7.8 s		M=4.23		h=	15.6 km		v=	1249 m/s		m=5.46e+05 kg

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101 RK t= 9.8 s | M=4.29 | h= 16.0 km | v=1265 m/s | m=5.42e+05 kg
102 RK t= 11.8 s | M=4.34 | h= 16.6 km | v=1281 m/s | m=5.38e+05 kg
103 RK t= 13.8 s | M=4.39 | h= 17.2 km | v=1297 m/s | m=5.34e+05 kg
104 RK t= 15.9 s | M=4.45 | h= 17.8 km | v=1313 m/s | m=5.29e+05 kg
105 RK t= 18.5 s | M=4.52 | h= 18.5 km | v=1334 m/s | m=5.24e+05 kg
106 RK t= 21.1 s | M=4.60 | h= 19.3 km | v=1357 m/s | m=5.19e+05 kg
107 RK t= 23.1 s | M=4.66 | h= 19.8 km | v=1375 m/s | m=5.15e+05 kg
108 RK t= 25.1 s | M=4.72 | h= 20.4 km | v=1392 m/s | m=5.11e+05 kg
109 RK t= 27.1 s | M=4.78 | h= 21.0 km | v=1410 m/s | m=5.06e+05 kg
110 RK t= 29.1 s | M=4.84 | h= 21.6 km | v=1429 m/s | m=5.02e+05 kg
111 RK t= 31.1 s | M=4.90 | h= 22.2 km | v=1447 m/s | m=4.98e+05 kg
112 RK t= 33.1 s | M=4.97 | h= 22.8 km | v=1466 m/s | m=4.94e+05 kg
113 RK t= 35.1 s | M=5.03 | h= 23.4 km | v=1485 m/s | m=4.90e+05 kg
114 RK t= 37.1 s | M=5.10 | h= 24.0 km | v=1504 m/s | m=4.86e+05 kg
115 RK t= 39.1 s | M=5.17 | h= 24.7 km | v=1524 m/s | m=4.82e+05 kg
116 RK t= 41.2 s | M=5.23 | h= 25.3 km | v=1545 m/s | m=4.78e+05 kg
117 RK t= 43.5 s | M=5.29 | h= 25.9 km | v=1569 m/s | m=4.73e+05 kg
118 RK t= 46.2 s | M=5.35 | h= 26.6 km | v=1597 m/s | m=4.68e+05 kg
119 RK t= 49.1 s | M=5.43 | h= 27.4 km | v=1629 m/s | m=4.62e+05 kg
120 RK t= 51.3 s | M=5.49 | h= 28.0 km | v=1654 m/s | m=4.57e+05 kg
121 RK t= 53.3 s | M=5.55 | h= 28.4 km | v=1677 m/s | m=4.53e+05 kg
122 RK t= 55.4 s | M=5.62 | h= 28.8 km | v=1702 m/s | m=4.49e+05 kg
123 RK t= 59.2 s | M=5.76 | h= 29.3 km | v=1749 m/s | m=4.41e+05 kg
124 RK t= 61.7 s | M=5.85 | h= 29.6 km | v=1780 m/s | m=4.36e+05 kg
125 RK t= 64.2 s | M=5.95 | h= 29.9 km | v=1813 m/s | m=4.31e+05 kg
126 RK t= 66.4 s | M=6.04 | h= 30.0 km | v=1843 m/s | m=4.26e+05 kg
127
128 === MISSION SUMMARY ===
129 Initial mass : 6.500e+05 kg
130 Final mass : 4.263e+05 kg
131 Propellant consumed : 2.237e+05 kg
132 1. Turbofan + AB : 4.612e+04 kg
133 2. Ramjet : 4.208e+04 kg
134 3. Rocket : 1.355e+05 kg
135 Payload fraction : 0.0231
136 Max Mach : 6.04 at 30.00 km
137 Total mission time : 549.2 s
138 Rocket burn duration: 66.4 s
139
140 --- Impulse Partition ---
141 Turbofan + AB : 9.05e+08 N.s (50.3%)
142 Ramjet : 4.95e+08 N.s (27.5%)
143 Rocket : 3.99e+08 N.s (22.2%)
144 Air-breathing : 77.8%
145
146 === REQUIREMENT VERIFICATION ===
147 Release Mach : 6.04 PASS
148 Release altitude : 30.0 km PASS
149 GTOW : 6.500e+05 kg PASS
150 Payload fraction : 0.0231 PASS
151 Max air-breath Mach : 4.00 PASS

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152	Air-breath impulse	:	77.8%	PASS
153	T04 max	:	2200 K	PASS
154	Rocket Isp	:	300 s	PASS
155	Rocket burn duration:	:	66.4 s	PASS